Institute of Geomatics, GIS and Remote Sensing

Integrating 3D Web-GIS and BIM Framework for Campus Indoor Routing Using Dijkstra's Algorithm And Cesium JS

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Background

- Modern institutional campuses have transitioned from horizontal layouts to dense, complex, multi-level vertical structures.
- High-density indoor environments make manual navigation difficult for visitors, new students, and emergency first responders.
- Outdoor pedestrian navigation is highly mature, relying on standard 2D network models and satellite positioning.
- Standard GPS signals fail inside buildings, creating a total spatial data blind spot the moment a user steps indoors.
- Traditional 2D GIS networks are mathematically incapable of calculating vertical routing transitions like multi-floor stairs, escalators, and elevators.
- Accurate indoor pathfinding requires explicit **3D network topology** to account for vertical connectivity (Z-coordinate routing).
- Highly detailed 3D geometric data exists inside architectural **Building Information Modeling (BIM)** files.

Problem Statement

- BIM(Architectural) data and GIS Traditionally exists in disconnected in isolated systems.
- Campus geospatial routing stops entirely at the building threshold.
- Existing web maps operate strictly on **2D planar networks**, making them algorithmically blind to vertical elevations Z-coordinates.
- Traditional outdoor routing engines cannot calculate routing through vertical connectors like multi-level stairs, elevators, or ramps.
- Without a unified **3D spatial network**, optimal pathfinding inside complex architectural structures is mathematically impossible.

Objectives

Overall objective

-To Integrate 3D Web-GIS and BIM Framework for Campus Indoor Routing Using Dijkstra's algorithm and Cesium JS.

Specific objectives

i. To evaluate and optimize the interoperability pipeline for transforming high-fidelity ArchiCAD BIM datasets into web-compatible 3D Tiles.

ii. To design a topological 3D spatial database schema that accurately preserves multi-floor building connectivity and network relationships.

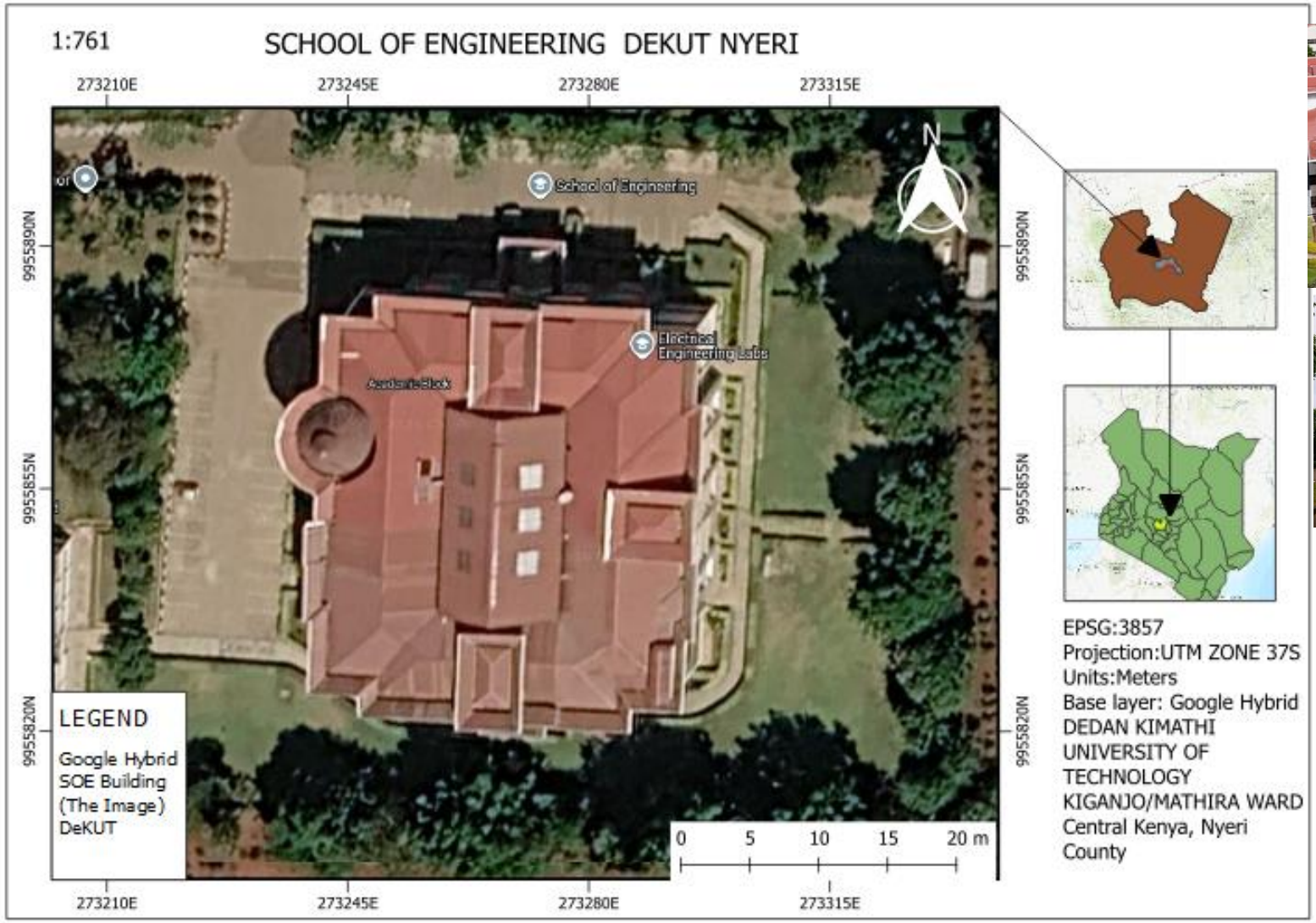
iii. To implement Dijkstra's algorithm within a Web-GIS environment.

Justification

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- **First Responders:** Accelerated crisis response via multi-floor vertical pathfinding.
- **Campus Community:** Reduced navigation confusion through browser-based 3D routing.
- **Mobility-Impaired Users:** Custom accessible routing by weighting Dijkstra's algorithm for elevators/ramps.
- **GIS Department:** A scalable blueprint for integrating BIM architectural data into Web-GIS.
- Routing based on 3D building environments improves campus usability and facility management efficiency.

Study Area

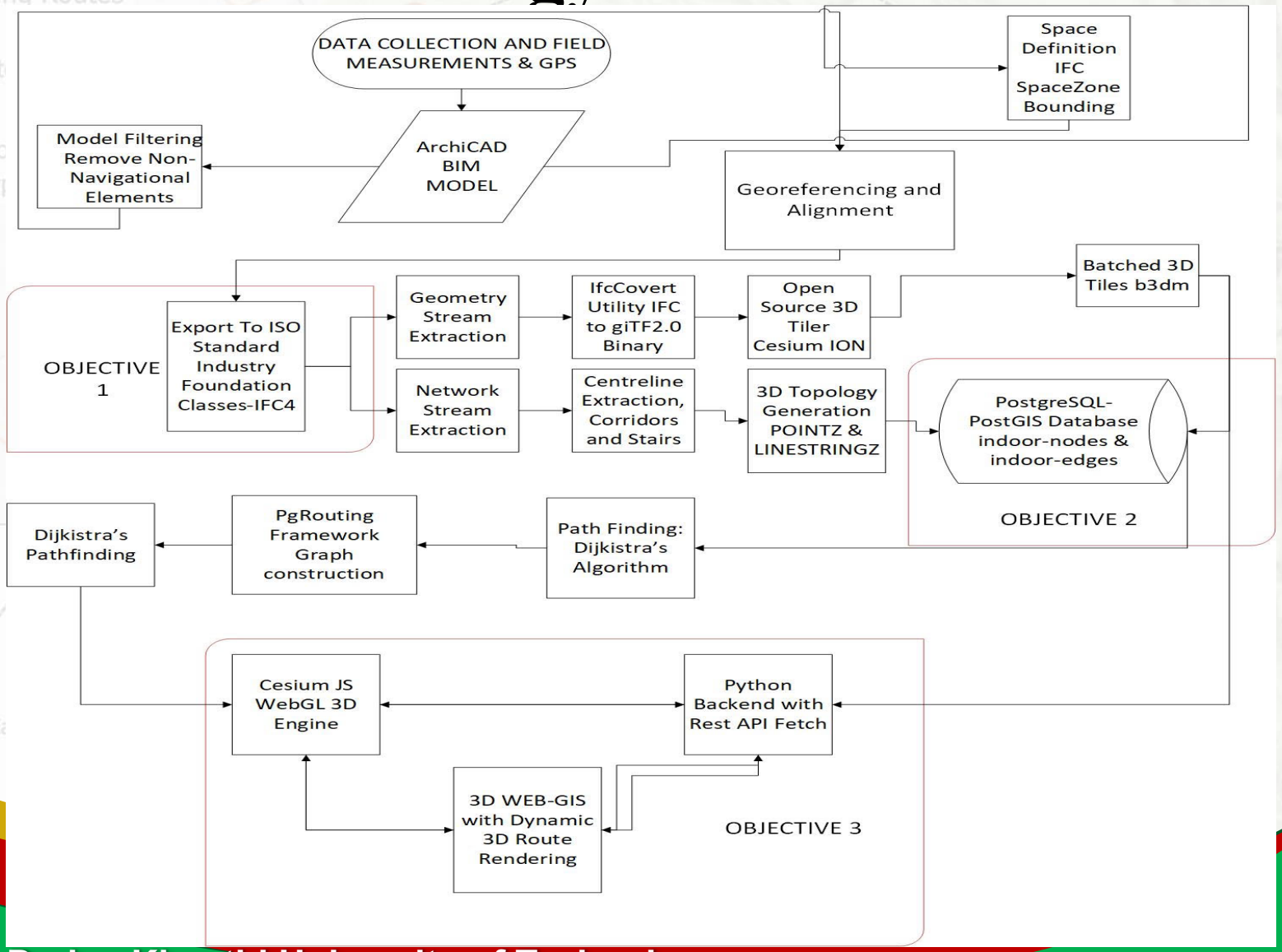

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Projection:UTM ZONE 37S
Units:Meters
Base layer: Google Hybrid
DEDAN KIMATHI
UNIVERSITY OF
TECHNOLOGY
KIGANJO/MATHIRA WARD
Central Kenya, Nyeri
County







Methodology: Flow Chart



Methodology

- Field Measurements: 1. Building dimensions 2. Sample coordinates for Georeferencing purpose using GPS.
- BIM model. ArchiCAD model(To model SOE floor plan using ArchiCAD)

Software and Tools

1. ArchiCAD.
2. Programming languages (Python and Java-script for Backend and pipeline.)
3. PostgreSQL with Post-GIS, and PgRouting Extensions.
4. Visual Studio Code.
5. Cesium Ion and Cesium js for 3D Models
6. Dijkstra's Algorithm for short path finding.

Expected Results

- An optimized 3D Tiles representation of campus building complexes streamed smoothly over standard web browsers via Cesium JS.
- A functional relational PostgreSQL/Post-GIS spatial database organizing architectural elements linked with live attributes.
- Indoor routing application that calculates and overlays the absolute shortest interior path between multi-floor rooms on the 3D model.


Legend
Mountain Climbing Routes
Kabaru Route
Burguret Route
GPS Waypoints
Kabaru Waypoints [38]
Burguret Waypoints [49]
Burguret Route
Kieni East
Kabaru Route
Kieni East
Nyeri Mt. Ker
Kieni West
Nyeri Municipality
Mathira
Tetu
2.5
0
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5
7.5
10 km

THANK YOU